

An Introduction to Eremenko's Conjecture on Complex Dynamic Systems

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Introduction and preliminaries

The dynamical study of transcendental entire functions was initiated by Fatou in 1926. Baker, Bergweiler, Kotus, Eremenko, Lyubich, McMullen, Rempe, Rippon, Stallard, Lv Yinian, Ren Fuyao, Fang Liping, Liao Liangwen, Qiao Jianyong, Qiu Weiyuan, Wang Yuefei, Wu Shengjian, Ng, Tuen Wai, Yang C.C., Zhang Gaofei, Zheng Jianhua etc. have done many works.

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Consider a transcendental function

$$f : \mathbb{C} \rightarrow \mathbb{C}.$$

The **Fatou set** $F(f)$ of f is defined as the maximal open subset of $\mathbb{C}$ in which the sequence $\{f^{\circ n}\}_{n \in \mathbb{N}}$ is equicontinuous. The **Julia set** of f is

$$J(f) = \mathbb{C} \setminus F(f).$$

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$$J(f) = \mathbb{C} \setminus F(f).$$

- $F(f)$ is open and $J(f)$ is closed.
- $F(f)$ and $J(f)$ are completely invariant.
- $F(f) = F(f^{\circ n})$ and $J(f) = J(f^{\circ n})$.

Definition

A point $s \in \mathbb{C}$ is called a singularity of the inverse function f^{-1} if s is either a critical value (the image of a critical point) or a finite asymptotic value. The latter means that there is a path to infinity whose image ends at s ; the curve is then called an asymptotic curve. The set of such singularities is denoted by $\text{sing}(f^{-1})$. Then we call

$$\mathcal{B} = \{f : \mathbb{C} \rightarrow \mathbb{C} \text{ transcendental entire} : \text{sing}(f^{-1}) \text{ is bounded}\}$$

the Eremenko-Lyubich class. The postsingular set is

$$\mathcal{P}(f) = \overline{\bigcup_{j \geq 0} f^j(\text{sing}(f^{-1}))}$$

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- ⑤ a preperiodic component that eventually maps to a periodic component in one of the types above;
- ⑥ a non-periodic component for which all forward iterates are disjoint (wandering domain).

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The first construction of wandering domain is due to I.N.Baker with the technique of infinite product in 1963:

$$f(z) = Cz^2 \prod_{j=1}^{\infty} \left(1 + \frac{z}{r_j}\right)$$

where $C > 0$ is a small constant, r_1 is large and r_n is sequence of positive numbers satisfies the recurrence relation:

$$r_{n+1} = Cr_n^2 \prod_{j=1}^n \left(1 + \frac{r_n}{r_j}\right)$$

Theorem (Baker, 1984)

If U is a multiply connected Fatou component of a transcendental entire function, then

- (1) U is a wandering domain in $I(f)$,*
- (2) $f^{n+1}(U)$ surrounds $f^n(U)$ for large n ,*
- (3) $\text{dist}(O, U_n) \rightarrow \infty$ as $n \rightarrow \infty$.*

Remark

Baker, Hinkkanen, Kisaka+Shishikura, Bergweiler, Osborne, Bishop.

Recently, Bishop gives a method which is called "Quasiconformal folding" to construct transcendental entire functions with good control of both the singular values of f and the geometry of the level-set $\{z : |f(z)| > R\}$.

Theorem (Bishop, Acta Math. 2015)

There is an $f \in \mathcal{B}$ whose Fatou set contains a wandering domain.

Remark

Baker's example does not belong to $\mathcal{B}$.

Theorem (Bishop, 2011)

There is an f with $\dim_H(J(f)) = 1$.

Remark

For $f \in \mathcal{B}$, $\dim_H(J(f)) > 1$.

History of Eremenko's conjecture

In his seminal 1926 article, Fatou observed that the Julia sets of certain explicit entire functions, such as $z \mapsto r \sin(z)$, $r \in \mathbb{R}$, contain curves of points that escape to infinity under iteration.

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History of Eremenko's conjecture

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He then remarked "It would be interesting to investigate whether this property might not hold for much more general functions."

History of Eremenko's conjecture

In 1989, Eremenko was the first to undertake a thorough study of the escaping set

$$I(f) := \{z \in \mathbb{C} : |f^n(z)| \rightarrow \infty\}$$

of an arbitrary transcendental entire function.

He showed that

$$I(f) \neq \emptyset,$$

$$\partial I(f) = J(f), I(f) \cap J(f) \neq \emptyset,$$

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Remark

Eremenko's construction of points in $I(f)$ is based on Wiman-Valiron theory.

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Remark

This can be seen as making Fatou's original question more precise, and will be referred to in the following as the **strong form of Eremenko's conjecture**.

Remark

For a meromorphic function f the set $I(f)$ was first considered by Dominguez in 1998. She proved that again $I(f) \neq \emptyset$, $\partial I(f) = J(f)$ and $I(f) \cap J(f) \neq \emptyset$. But in this situation $\overline{I(f)}$ need not have unbounded components. For $f(z) = \frac{1}{2} \tan z$, $\overline{I(f)} = J(f)$, and this set is totally disconnected.

Previous results before 1989

- 1 In 1984, Devaney gave a complete description of the Julia set of any real exponential map that has an attracting fixed point; that is, $z \mapsto \lambda \exp(z)$ with $\lambda \in (0, \frac{1}{e})$. This was the first entire function for which it was discovered that the escaping set (and in fact the Julia set) consists of curves to ∞ .

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Remark

It seems that it was partly these developments that led Eremenko to pose the above-mentioned questions in his 1989 paper.

Previous results after 1989

- 1 In 2003, D. Schleicher and J. Zimmer proved that every escaping point of every exponential map can be connected to ∞ by a curve consisting of escaping points. This was the first time that a complete classification of all escaping points, and with it a positive answer to both of Eremenko's questions, was given for a complete parameter space of transcendental functions.
- 2 In 2008, Rottenfusser and D. Schleicher carried this result over to the cosine family $z \mapsto a \exp(z) + b \exp(-z)$.

Partial answers of Eremenko's conjecture

Theorem (Rippon, Stallard, 2005, Proc. Amer. Math. Soc.)

Let f be a transcendental entire function. Then each $z_0 \in A(f)$ lies in an unbounded closed connected subset of $A(f)$; in particular, $A(f)$ has no bounded components.

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Remark

This result was proved by considering the fast escaping set $A(f)$,

$$A(f) = \{z : \text{there exists } l \in \mathbb{N} \text{ such that } |f^{n+l}(z)| > M(R, f^n) \text{ for } n \in \mathbb{N}\},$$

which was introduced by Bergweiler and Hinkkanen in 1999 and which now plays a key role in transcendental dynamics.

Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be a transcendental entire function and, for $r > 0$, let $z(r)$ denote a point such that

$$|z(r)| = r \quad \text{and} \quad |f(z(r))| = M(r).$$

Also, let $N(r)$ be the largest value of n for which $|a_n| r^n$ is maximal. Note that $N(r)$ is increasing with r and $N(r) \rightarrow \infty$ as $r \rightarrow \infty$.

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A key result of Wiman-Valiron theory is the following.

Theorem

Let f be a transcendental entire function, and let $\alpha > \frac{1}{2}$ and $r > 0$. There exists a set $E \subset (0, \infty)$ such that $\int_E \frac{1}{t} dt < \infty$ and, for $r \notin E$, if $|z - z(r)| < \frac{r}{(N(r))^\alpha}$, then

$$f(z) = \left(\frac{z}{z(r)}\right)^{N(r)} f(z(r))(1 + \varepsilon),$$

where $\varepsilon = \varepsilon(r, z) \rightarrow 0$ uniformly with respect to z as $r \rightarrow \infty, r \notin E$.

Eremenko used the above theorem to deduce a result of the following type.

Theorem (Rippon, Stallard, 2012, Proc. London Math. Soc.)

Let f be a transcendental entire function, and let $K > 1$ and $r > 0$. There exist $C(K) > 0$ and a set $E_K \subset (0, \infty)$ with $\int_E \frac{1}{t} dt < \infty$ such that, if $D_{r,K} = \{z : |z - z(r)| < \frac{C(K)r}{N(r)}\}$, then, for $r \notin E_K$,

$$f(D_{r,K}) \supset A_{r,K} = \{z : \frac{1}{K}M(r) \leq |z| \leq KM(r)\},$$

and, if B is a compact subset of a disc $D \subset A_{r,K}$ and B_{-1} is a component of $D_{r,K} \cap f^{-1}(B)$, then

$$\text{diam} B \geq c(K)N(r)\frac{M(r)}{r}\text{diam} B_{-1},$$

where $c(K) > 0$ is a constant that depends only on f and K .

Rippon and Stallard used the above theorem to obtain the points in $A(f)$ which have other strong dynamical properties.

Theorem (Rippon, Stallard, 2012, Proc. London Math. Soc.)

Let f be a transcendental entire function, and let $\varepsilon \in (0, 1)$ and $K \geq 2$. There exist $R > 0$ such that, if $r > R$, then there exists

$$z' \in \{z : r \leq |z| \leq r(1 + \varepsilon)\}$$

with

$$|f^n(z')| > M^n(r) \text{ for } n \in \mathbb{N},$$

and hence $z' \in A_R(f) = \{z : |f^n(z)| \geq M^n(R, f) \text{ for } n \in \mathbb{N}\} \subset A(f)$.

① Bergweiler, Hinkkanen (1999, Math. Proc. Cambridge Philos. Soc.).

$$A'(f) = \{z : \text{there exists } l \in \mathbb{N} \text{ such that } |f^{n+l}(z)| > M(R, f^n) \text{ for } n \in \mathbb{N}\},$$

where R is any value such that $R > \min_{z \in J(f)} |z|$.

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$$B(f) = \{z : \text{there exists } l \in \mathbb{N} \text{ such that } f^{n+l}(z) \notin \widetilde{f^n(D)} \text{ for } n \in \mathbb{N}\},$$

where D is any open disc meeting $J(f)$, and $\widetilde{U}$ denotes the union of U and its bounded complementary components.

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Theorem (Rippon and Stallard (2012, Proc. London Math. Soc.))

Let f be a transcendental entire function. Then

$$A(f) = A'(f) = B(f).$$

Theorem (Rempe, 2007, Bull. London Math. Soc.)

Let $f \in \mathcal{B}$ be such that $\mathcal{P}(f) = \overline{\bigcup_{j \geq 0} f^j(\text{sing}(f^{-1}))}$ is bounded. Then every component of $I(f)$ is unbounded.

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Theorem (Gunter Rottenfuer, Johannes Ruckert, Lasse Rempe, and Dierk Schleicher, 2011, Ann. Math.)

There exists a hyperbolic entire function $f \in \mathcal{B}$ such that every path-connected component of $I(f)$ is bounded. On the other hand, for many functions in $\mathcal{B}$, in particular those of finite order, every escaping point can be connected to ∞ by a curve of escaping points.

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Remark

The techniques in [Bishop, C. J., Constructing entire functions by quasiconformal folding, Acta Math., 2015] show that the above example can be replicated in $\mathcal{S}$.

Theorem (Rempe,Rippon,Stallard,2009,J of Diff.Equ.App.)

Let f be a finite composition of functions of finite order in the class $\mathcal{B}$ and let $z_0 \in I(f)$. Then z_0 can be connected to ∞ by a simple curve $\gamma \subset I(f)$ such that

- (a) $f^n \rightarrow \infty$ as $n \rightarrow \infty$ uniformly on γ ,*
- (b) $\gamma \setminus \{z_0\} \subset A(f)$.*

Answered by using spider's web

Definition (Rippon, Stallard, Proc. London Math. Soc., 2012)

A set E to be a spider's web if E is connected, and there exists a sequence $(G_k)_{k \in \mathbb{N}}$ of bounded, simply connected domains such that $G_k \subset G_{k+1}$ and $\partial G_k \subset E$, for each $k \in \mathbb{N}$, and $\bigcup_{k \in \mathbb{N}} G_k = \mathbb{C}$.

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If $I(f)$ is a spider's web, then $I(f)$ is connected (and unbounded) and so Eremenko's conjecture holds.

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Theorem (Rippon, Stallard, Proc. London Math. Soc., 2012)

Let f be a transcendental entire function and let $R > 0$ be such that $M(r, f) > r$ for $r \geq R$. If $A_R(f)^c$ has a bounded component, then each of $A_R(f)$, $A(f)$ and $I(f)$ is a spider's web.

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Let f be a transcendental entire function, let $R > 0$ be such that $M(r, f) > r$ for $r \geq R$ and let $A_R(f)$ be a spider's web.

(a) If f has no multiply connected Fatou components, then each of

$$A_R(f) \cap J(f), A(f) \cap J(f), I(f) \cap J(f), \text{ and } J(f)$$

is a spider's web.

(b) The function f has no unbounded Fatou components.

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Baker's conjecture(1981): $F(f)$ has no unbounded components whenever the order of the function is less than $\frac{1}{2}$ or even whenever the function has order at most $\frac{1}{2}$ minimal type.

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Definition

The order of growth ρ of a transcendental entire function f is defined by $\rho = \limsup_{r \rightarrow \infty} \frac{\log \log M(r, f)}{\log r}$, where $M(r) = \max_{|z|=r} |f(z)|$.

Define σ by $\sigma = \limsup_{r \rightarrow \infty} \frac{\log M(r)}{r^\rho}$, we say that the growth of f is of minimal type if $\sigma = 0$, mean type if $0 < \sigma < \infty$ and maximal type if $\sigma = \infty$.

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Remark

It was proved by Stallard(1991), Qiao Jianyong(1994) and Anderson and Hinkkanen(1998) that every preperiodic or periodic component of $F(f)$ is bounded provided that f is of order less than $\frac{1}{2}$.

Theorem (Zheng,Jianhua,2000,Math. Proc. Camb. Phil. Soc.)

Let f be a transcendental entire function of growth not exceeding order $\frac{1}{2}$, minimal type. Then every preperiodic or periodic component of $N(f)$ is bounded.

Theorem (Wang Yuefei,2001,Israel J. Math.)

Let f be an entire function of order less than $1/2$. Then if its lower order $\mu > 0$, every component of $F(f)$ is bounded.

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Let f be a transcendental entire function of growth not exceeding order $\frac{1}{2}$, minimal type. Then every preperiodic or periodic component of $N(f)$ is bounded.

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Let f be an entire function of order less than $1/2$. Then if its lower order $\mu > 0$, every component of $F(f)$ is bounded.

Definition

Suppose that $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is an entire function with gaps, i.e., such that many of the a_n are zero, in a certain sense. Then the function has the form $f(z) = \sum_{k=0}^{\infty} a_k z^{n_k}$. We say that $f(z)$ has Fabry gaps if $\frac{n_k}{k} \rightarrow \infty$.

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Theorem (Wang Yuefei, 2001, Tohoku Math. J.)

Let $f(z) = \sum_{k=0}^{\infty} a_k z^{n_k}$ be an entire function with $0 < \mu \leq \rho < \infty$. If $f(z)$ has Fabry gaps, then every component of $F(f)$ is bounded.

Theorem (Rippon, Stallard, Proc. London Math. Soc., 2012)

Let f be a transcendental entire function and let $R > 0$ be such that $M(r, f) > r$ for $r \geq R$. Then $A_R(f)$ is a spider's web if one of the following holds:

- (a) f has a multiply connected Fatou component;
- (b) f has very small growth, that is, there exist $m \geq 2$ and $r_0 > 0$ such that

$$\log \log M(r, f) < \frac{\log r}{\log^m r} \text{ for } r > r_0,$$

where $\log^m$ denotes the m th iterated logarithm function;

- (c) f has order less than $1/2$ and regular growth;
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Remark

Both Eremenko's conjecture and Baker's conjecture hold for such functions.

Definition (Rippon, Stallard, 2013, Inter.Math. Research Notices)

For $\mu_\varepsilon(r) = M(r)^\varepsilon$, define

$$Q_\varepsilon(f) = \{z : \text{there exists } l \in \mathbb{N} \text{ such that } |f^{n+l}(z)| \geq \mu_\varepsilon^n(R), \text{ for } n \in \mathbb{N}\},$$

where $R > 0$ has the property that $\mu_\varepsilon(r) > r$ for $r \geq R$. It is easy to check that the definition of $Q_\varepsilon(f)$ is independent of the choice of R with the property that $\mu_\varepsilon(r) > r$ for $r \geq R$. Define $Q(f) = \bigcup_{0 < \varepsilon < 1} Q_\varepsilon(f)$ to be the quite fast escaping set of f .

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Theorem (Rippon, Stallard, 2013, Inter.Math.Research Notices)

Let $f \in \mathcal{B}$ be a transcendental entire function. Then $Q(f) = A(f)$.

Meromorphic functions with direct tracts

Based on the analysis of the definition of the direct tracts, Bergweiler, Rippon and Stallard in 2008 used the subharmonic function theory to prove the following Wiman-Valiron type theory coming from entire functions (we denote it by Wiman-Valiron-BRS theory)

Definition (Bergweiler, Rippon, Stallard, 2008, Proc. London Math. Soc.)

Let D be an unbounded domain in $\mathbb{C}$ whose boundary consists of piecewise smooth curves. Suppose that the complement of D is unbounded. Let f be a complex-valued function whose domain of definition contains the closure $\overline{D}$ of D . Then D is called a direct tract of f if f is holomorphic in D and continuous in $\overline{D}$ and if there exists $R > 0$ such that $|f(z)| = R$ for $z \in \partial D$ while $|f(z)| > R$ for $z \in D$. If, in addition, the restriction $f : D \rightarrow \{z \in \mathbb{C} : |z| > R\}$ is a universal covering, then D is called a logarithmic tract of f .

Note that if f, D, R are as in the above definition, then the function $v : \mathbb{C} \rightarrow [0, \infty)$ defined by $v(z) = \begin{cases} \log \frac{|f(z)|}{R} & , \text{if } z \in D, \\ 0 & , \text{if } z \notin D \end{cases}$ is subharmonic.

Then $B(r, v) = \max_{|z|=r} v(z)$ is increasing, convex in $\log r$ and tends to ∞ as r tends to ∞ . Hence $a(r, v) = \frac{dB(r, v)}{d \log r} = rB'(r, v)$ exists except, perhaps, for a countable set of r -values, and $a(r, v)$ is non-decreasing. We also put $M_D(r) = \max_{|z|=r, z \in D} |f(z)| = \exp B(r, v)$. Define

$$I(f, D) = \{z \in D : f^n(z) \in D, f^n(z) \rightarrow \infty \text{ as } n \rightarrow \infty\}$$

$$A(f, D, r) = \{z \in D : f^n(z) \in D \text{ for all } n \in \mathbb{N} \mid |f^n(z)| \geq M_D^n(r)\}.$$

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Remark

We ask the following:

Eremenko's conjecture for the meromorphic functions with at least one direct tract:

Is every component of $I(f, D)$ unbounded?

Theorem (Bergweiler, Rippon, Stallard, 2008, Proc. London Math. Soc.)

Let D be a direct tract of f and let $\tau > \frac{1}{2}$. Let $v(z)$ be defined as before and let z_r be a point satisfying $|z_r| = r$ and $v(z_r) = B(r, v)$. Then there exists a set $F \subset [1, +\infty)$ of finite logarithmic measure such that if, $r \in [1, +\infty) \setminus F$, then $D\left(z_r, \frac{r}{a(r, v)^\tau}\right) \subset D$,

$$f(z) \sim \left(\frac{z}{z_r}\right)^{a(r, v)} f(z_r), \quad z \in D\left(z_r, \frac{r}{a(r, v)^\tau}\right) \quad (1)$$

and

$$|f(z)| \sim M_D(|z|), \quad z \in D\left(z_r, \frac{r}{a(r, v)^\tau}\right). \quad (2)$$

as $r \rightarrow \infty, r \notin F$.

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Remark

This is the key result of the Wiman-Valiron-BRS-type theory.

Definition (Bergweiler, Rippon, Stallard, 2008, Proc. London Math. Soc.)

Let f be a transcendental function meromorphic in the plane which has a direct tract D . R is the number, define

$$\begin{aligned} A(f, D) &= \{z \in \mathbb{C} : \exists K \in \mathbb{N} \text{ such that } f^K(z) \in A(f, D, R)\} \\ &= \{z \in \mathbb{C} : \exists K \in \mathbb{N} \text{ such that } f^{n+K}(z) \in D \text{ for all } n \in \mathbb{N}, \\ &\quad \text{and } |f^{n+K}(z)| \geq M_D^n(R)\}. \end{aligned} \tag{3}$$

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Definition (Xuan,Zheng,2015)

The L th level of $A(f, D)$ (with respect to R) is the set

$$A_R^L(f, D) = \left\{ z : f^n(z) \in D, |f^n(z)| \geq M_D^{n+L}(R), n \in \mathbb{N}, n+L \geq 0 \right\}.$$

Theorem (Xuan,2024,Preprint)

Define

$$A'(f,D) = \left\{ z : \exists l \in \mathbb{N} \text{ such that for } n \in \mathbb{N}, \text{ we have } |f^{n+l}(z)| \geq M_D(R, f^n) \right\}$$

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$$A(f,D) = A'(f,D).$$

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Then

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Definition (Xuan,2024,Preprint)

For $\mu_\varepsilon(r, D) = M_D(r)^\varepsilon$, define

$$Q_\varepsilon(f, D) = \{ z : \text{there exists } l \in \mathbb{N} \text{ such that } |f^{n+l}(z)| \geq \mu_\varepsilon^n(R, D), \text{ for } n \in \mathbb{N} \},$$

where $R > 0$ has the property that $\mu_\varepsilon(r) > r$ for $r \geq R$. It is easy to check that the definition of $Q_\varepsilon(f, D)$ is independent of the choice of R with the property that $\mu_\varepsilon(r) > r$ for $r \geq R$.

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Remark

It follows immediately from these definitions that

$$A(f, D) \subset Q_\varepsilon(f, D) \subset Q(f, D) \subset I(f, D) \text{ for } 0 < \varepsilon < 1.$$

Theorem (Xuan,2024,Preprint)

Let f be a meromorphic function with a logarithmic singularity over infinity. Then

$$Q(f, D) = A(f, D).$$

THANK YOU FOR YOUR ATTENTION!